



Original Research

Decreased ankle and hip isometric peak torque in young and middle-aged adults with chronic ankle instability



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ABSTRACT

Objectives: To compare ankle, knee and hip isometric peak torque between young and middle-aged adults with CAI, copers and un-injured controls.

Design: Cross-sectional.

Setting: Research Laboratory.

Participants: One hundred fifty-six young and middle-aged adults with or without CAI volunteered.

Main outcome measures: A handheld dynamometer measured isometric dorsiflexion, plantarflexion, knee extension, hip extension and hip abduction peak force during a 5 s trial. Average peak torque was calculated and normalized to body mass.

Results: A significant Age by Injury interaction for dorsiflexion suggest middle-aged un-injured controls ($p < 0.001$) and copers ($p < 0.001$) had lower isometric peak torque compared to their young adult counterparts, but there were no differences between young and middle-aged adults with CAI ($p > 0.05$). Significant Injury main effects suggest the CAI group had decreased plantarflexion ($p = 0.004$) and hip extension ($p = 0.010$) strength compared to un-injured controls, but not copers ($p > 0.05$). Significant Age main effects for all primary outcome measures were observed, indicating peak torque decreased with age ($p < 0.05$).

Conclusions: Regardless of age, isometric ankle and hip peak torque was lower in participants with CAI compared to un-injured controls, but not copers. These findings provide further evidence towards the impact of CAI in both young and middle-aged adults.

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1. Introduction

Ankle sprains account for 7% to 15% of all U.S. collegiate athletic injuries and result in more than 600,000 Emergency Department visits each year (Hootman, Dick, & Agel, 2007; Waterman, Owens, Davey, Zacchilli, & Belmont, 2010). There are two primary responses in the recovery of a lateral ankle sprain that are associated with a spectrum of clinical outcomes (Hertel & Corbett, 2019). Coper is the term normally given to individuals who return to their

regular physical activity with minimal dysfunction, no 'giving way' or pain (Wikstrom & Brown, 2014). In contrast, up to 73.6% of people do not fully recover and develop chronic ankle instability (CAI) (Anandacoomarasamy & Barnsley, 2005; Gerber, Williams, Scoville, Arciero, & Taylor, 1998). CAI develops when individuals have recurrent ankle sprains, repeated episodes of 'giving way' and perceived ankle joint instability (Doherty et al., 2016; Gribble et al., 2014). The high rate of re-injury and persistent symptoms can significantly impact a person's health-related quality of life (Houston, Hoch, & Hoch, 2015) and the ability to remain physically active (Hubbard-Turner & Turner, 2015). Further, there is growing evidence linking CAI with early biomarkers of post-traumatic ankle osteoarthritis (Golditz et al., 2014; Wikstrom et al., 2019). This cascade of negative consequences associated with CAI underscores

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the need for continued research into the impairments that underly this condition. To better understand the impairments that contribute to this condition, researchers have encouraged the inclusion of copers into research studies because they appear to have been able to respond with no chronicity of negative effects following an ankle sprain.

An updated model describes CAI as a multifactorial clinical pathology arising from several pathomechanical, sensory-perceptual and motor-behavioral impairments (Hertel & Corbett, 2019). Lower extremity muscle weakness is one common motor-behavioral impairment that has been examined extensively among individuals with CAI (Hertel & Corbett, 2019). Inconsistent findings have caused the authors of systematic reviews to question whether frontal plane ankle strength is associated with CAI (Kaminski & Hartsell, 2002; Khalaj, Vicenzino, Heales, & Smith, 2020). In contrast, sagittal plane ankle strength has found to be diminished in between those with and without CAI (Fraser et al., 2018) and is a risk factor for an inversion ankle sprain (Baumhauer, Alosa, Renstrom, Trevino, & Beynnon, 1995; Fousekis, Tsepis, & Vagenas, 2012). Others have noted proximal strength deficits in the knee and hip musculature among individuals with CAI relative to un-injured controls and copers (Gribble & Robinson, 2009; McCann et al., 2017). Researchers have hypothesized different mechanisms contributing to these strength deficits including centrally-mediated neural alterations or abnormal movement patterns leading to disuse (DeJong, Koldenhoven, & Hertel, 2020; Fraser et al., 2018; Gribble & Robinson, 2009). While the exact mechanism is unknown, it is clear that reduced sagittal plane lower extremity strength is a concern among this population as deficiencies have been associated with decreased functional performance in tasks such as the Star Excursion Balance Test (Gabriner, Houston, Kirby, & Hoch, 2015; McCann et al., 2017) and landing kinematics (McCann, Terada, Kosik, & Gribble, 2019).

The research focused on understanding the progression, impairments, consequences and rehabilitation strategies for CAI has primarily included young adult populations (18–40 years of age) (Delahunt et al., 2010). This trend is likely the result of investigators relying on a convenient sample of college-aged participants (Delahunt et al., 2010). However, this approach limits the knowledge of whether middle-aged CAI participants present with similar impairments and if these impairments continue to decline at a greater rate compared to un-injured adults or copers. A population-based survey of Australian citizens (median age of 48 years) illustrated that the majority of respondents with chronic musculoskeletal ankle disorders had symptoms lasting longer than 10 years (Hiller et al., 2012). Simon and Docherty (Simon & Docherty, 2018) reported middle-aged adults with CAI display a lower health-related quality of life compared to their healthy counterpart, providing support towards the impact of CAI later in life. The results from these studies imply that the lingering symptoms after an ankle sprain can continue throughout life and are not limited to young adults. Despite the growing understanding of CAI beyond young adulthood, no investigation has examined lower extremity strength in middle-aged adults with CAI. This is important to document because muscle strength continues to decline with age and is a prominent contributor to physical performance later in life (Tieland, Trouwborst, & Clark, 2018). Therefore, the muscle weakness previously observed in young adults with CAI may pose a risk for an early decline in physical function in middle-aged adults. It is also possible that muscle weakness may be a contributing factor to the lower health-related quality of life documented in middle-aged adults with CAI (Simon & Docherty, 2018) because it could be impacting their ability to remain physically active and participate in social activities.

The purpose of this study was to compare multi-joint strength

measured with isometric peak torque among young and middle-aged adults with CAI, copers and controls. We hypothesized middle-aged participants would have lower ankle, knee and hip isometric peak torque compared to young adults, regardless of injury history. Secondly, we hypothesized participants with CAI would have lower isometric ankle, knee and hip peak torque compared to copers and un-injured controls, regardless of age. Thirdly, we hypothesized middle-aged participants with CAI would have lower isometric ankle, knee and hip peak torque compared to young-adults in all three injury groups, and middle-aged copers and un-injured controls.

2. Methods

2.1. Experimental approach to the problem

A laboratory-based, descriptive cross-sectional study design was employed to compare ankle, knee and hip isometric peak torque between young and middle-aged adults with CAI, copers and controls. All participants reported to the research laboratory within two weeks of screening for a single testing session. The primary investigator was responsible for recruitment, enrollment, data collection and analysis.

2.2. Participants

One hundred fifty-six participants between 18 and 70 years of age from a university and its surrounding community from 2016 to 2018 volunteered. The primary aim and experimental procedures were explained to the participants prior to enrollment. All participants were required to be free from any: 1) diagnosed balance, vestibular or respiratory disorder; 2) history of low back pain or a concussion in the 6 months prior to enrollment; 3) previous history of lower extremity fracture or surgery; 4) history of seizures; 5) history of neurological injuries or diseases; and 6) current use of an assisted walking device. All participants read and signed the informed consent approved by the institutional review board (#15–0966).

Previously published data examining the health-related quality of life in middle-aged adults with and without CAI was used to generate age classifications for the current study (Simon & Docherty, 2018). A young adult fell between 18 and 40 years of age ($n = 99$) and a middle-aged adult fell between 41 and 70 years of age ($n = 57$).

The inclusion criteria for participants with CAI followed the guidelines suggested by the International Ankle Consortium (Gribble et al., 2014). Participants allocated to the CAI group had to: 1) have an acute ankle sprain that occurred more than 12 months before enrollment that resulted in swelling, pain and temporary loss of function for at least 1 day; 2) have at least two episodes of 'giving way' in the 6 months before enrollment; and 3) score >5 on the Ankle Instability Instrument and >11 on the Identification of Functional Ankle Instability Instrument (IdFAI). Inclusion criteria for participants in the copers group reflected the recommendations by Wikstrom & Brown (Wikstrom & Brown, 2014). Participants allocated to the copers group had to: 1) have an acute ankle sprain that occurred more than 12 months before enrollment that resulted in swelling, pain and temporary loss of function for at least 1 day; 2) indicate never experiencing a sensation of 'giving way'; and 3) score <5 and <11 on the AII and IdFAI, respectively. Additionally, all middle-aged participants with a previous history of an ankle sprain had to self-report their first ankle sprain occurred before 40 years of age. This was used as an inclusion criterion to try and examine the lasting effects of an ankle sprain and attempt to ensure participants had either developed CAI

or became a coper as a young adult.

In the event a participant had a history of bilateral ankle sprains, the involved limb was identified based on the greatest number of 'giving way' episodes and the highest score on the All/IdFAI. Participants allocated to the control group had to: 1) self-report having no history of a previous acute ankle sprain; 2) self-report having no repeated episodes of 'giving way'; and 3) have a score of 0 on the All and IdFAI. Participants were not matched for sex or injury limb.

2.3. Instrumentation

A BTE Evaluator (BTE™ Evaluator, BTE, Hanover, MD) with a portable loadcell with attachments designed for handheld dynamometer was used to measure ankle, knee and hip peak isometric force.

2.4. Experimental procedures

After reading and signing an informed consent approved by the institutional review board, participants completed a health-history survey, the All and IdFAI. The health-history survey included questions aimed at recording the number of years each participant experienced symptoms (e.g. giving-way or instability) related to CAI and contextual information such as if participants have to modify their activity because of their ankle, if they feel at risk of re-injury or if they received physical rehabilitation for an ankle sprain from a certified athletic trainer or physical therapist.

After completing the health-history survey, isometric dorsiflexion, plantar flexion, hip extension, hip abduction and knee extension force assessments were performed on the involved limb only and occurred in the same order for all participants. Participant positioning for each isometric force assessment was based on previously published data (Fig. 1) (Andrews, Thomas, & Bohannon, 1996; Ireland, Willson, Ballantyne, & Davis, 2003; Spink, Fotoohabadi, & Menz, 2010; Toonstra & Mattacola, 2013). Dorsiflexion and plantar flexion isometric force was measured with the participants lying on a standard plinth in a supine and prone position, respectively. The hip, knee and ankle were placed at 0° and two straps were secured over the upper and lower leg to prevent any accessory motion. The handheld dynamometer was placed over the metatarsal heads on the dorsum of the foot for dorsiflexion and on the plantar surface of the foot for plantar flexion. Isometric hip extension force was measured in a prone position with the hip at 0° and the knee flexed to 30°. A strap was secured over the hip to limit accessory motion and the handheld dynamometer was secured using a second strap just proximal to the popliteal fossa. Isometric hip abduction force was measured in a side-lying position with the involved limb facing upward and the hip, knee and ankle at 0°. A pillow was placed between both limbs and a strap was secured over the waist to limit accessory motion. The handheld

dynamometer was positioned 5.08 cm (2 in) proximal to the lateral joint line and secured using a second strap. Isometric knee extension force was measured with participants seated on the edge of a standard plinth with their hips and knees flexed to 90° and their arms resting across the chest. A strap was placed over the thighs to limit accessory motion. The portable load cell was secured behind the involved limb with one end attached to a stabilizing strap located approximately 5.08 cm (2 in) above the medial malleoli of the involved limb and the other tied around the plinth. Using these testing positions, good-to-excellent test-retest reliability for handheld dynamometry has been demonstrated for dorsiflexion (ICC = 0.91) (Spink et al., 2010), plantar flexion (ICC = 0.89) (Spink et al., 2010), hip abduction (ICC = 0.78) (Thorborg, Petersen, Magnusson, & Holmich, 2010), hip extension (ICC = 0.81) (Thorborg et al., 2010) and knee extension (ICC = 0.92) (Toonstra & Mattacola, 2013).

Participants were instructed to provide resistance for 5-s by slowly increasing force during the first 2-s and then providing maximal effort for the final 3-s. Each participant completed a practice trial and three test trials were recorded. A 30-s rest interval was given between trials.

2.5. Data reduction

Peak force was recorded in pounds (lbs.) and then converted to Newtons (N) and averaged across all three test trials. Isometric peak torque was calculated by taking the product of the average peak force (N) and the external moment arm (m). The external moment arm for dorsiflexion and plantar flexion was the linear distance from the posterior lateral malleolus to the metatarsal heads. The moment arm for hip abduction and hip extension was the linear distance between the greater trochanter and the lateral joint line of the knee. The external moment arm for knee extension was the linear distance between the lateral joint line of the knee and the most distal aspect of the lateral malleolus. Finally, the average isometric peak torque was normalized to each participant's body mass (Nm/kg) and used for statistical analysis.

2.6. Statistical analysis

The Kolmogorov-Smirnov test indicated all key demographic information and primary outcome measures were normally distributed ($p > 0.05$). Therefore, separate 3x2 between-group analysis of variance models were conducted to explore the impact of Injury (CAI, coper & control) and Age (younger & middle) on all key demographic information and primary outcome measures. A Tukey HSD post hoc analysis was used in the event of a significant interaction or Injury main effect. The *a priori* alpha level was set at $p < 0.05$. Mean difference and bias corrected Hedge's *g* effect sizes with their associated 95% confidence intervals (CI) were



A. Dorsiflexion; B. Plantarflexion; C. Hip Abduction; D. Hip Extension; E. Knee Extension

Fig. 1. Participant positioning for ankle, knee and hip isometric force assessments. A. Dorsiflexion; B. Plantarflexion; C. Hip Abduction; D. Hip Extension; E. Knee Extension

calculated for any group differences. All statistical analyses were conducted using IBM SPSS Statistics version 25.0 (IBM, Corp., Armonk, NY, USA).

3. Results

Group means and standard deviations or frequencies for the questionnaires used for inclusion criteria, injury history and participant characteristics can be found in Table 1.

Group means and standard deviations for all key demographic information are listed in Table 2. No significant Injury by Age interaction was observed for any key demographic information ($p > 0.05$). There was a significant main effect for Age, indicating middle-aged participants were older ($p < 0.001$) and had a higher mass ($p = 0.014$) and BMI ($p < 0.001$) than the young adults. Additionally, a significant main effect for Injury was observed for mass ($p = 0.014$) and BMI ($p = 0.007$). Post hoc analysis indicated that the CAI group had a higher mass ($p = 0.044$) and BMI ($p = 0.007$) compared to the control group. No significant differences were observed between the CAI and copers group or between the copers and control groups.

Group means and standard deviations for ankle, knee and hip isometric peak torque can be found in Table 3. A significant Injury by Age interaction was observed for dorsiflexion isometric peak torque ($p = 0.033$). Post-hoc testing showed that middle-aged controls (Hedge's g : 1.07, 95% CI[0.53, 1.61]; mean difference: 0.10; $p < 0.001$) and copers (Hedge's g : 1.66, 95% CI[0.95, 2.37]; mean difference: 0.16; $p < 0.001$) had lower dorsiflexion isometric peak torque compared to the young adult counterparts, but there were no differences between young and middle-aged adults with CAI ($p > 0.05$).

There was a significant main effect for Age, indicating middle-aged adults had decreased isometric plantar flexion (Hedge's g : 0.54, 95% CI[0.21, 0.87]; mean difference: 0.09; $p = 0.001$), knee extension (Hedge's g : 0.60, 95% CI[0.27, 0.93]; mean difference: 0.38; $p = 0.001$), hip abduction (Hedge's g : 0.52, 95% CI[0.19, 0.86]; mean difference: 0.27; $p = 0.002$) and hip extension (Hedge's g : 0.82, 95% CI[0.48, 1.16]; mean difference: 0.50; $p < 0.001$) peak torque compared to the young adults. Additionally, significant Injury main effects for plantar flexion ($p = 0.009$) and hip extension ($p = 0.018$) were observed. Post hoc analysis demonstrated that the CAI group had decreased isometric plantarflexion peak torque compared to the control (Hedge's g : 0.58, 95% CI[0.18, 0.99]; mean difference: 0.10; $p = 0.004$) group but not the copers ($p > 0.05$). Likewise, the CAI group had decreased isometric hip extension peak torque compared to the control (Hedge's g : 0.52, 95% CI[0.11, 0.92]; mean difference: 0.33; $p = 0.010$) group but not the copers ($p > 0.05$).

4. Discussion

Muscle strength plays a prominent role in preserving physical function later in life (Tieland et al., 2018). The known muscle weakness in young adults with CAI may increase the risk for an early decline in physical function or contribute to the lower health-related quality of life previously documented in middle-aged adults (Simon & Docherty, 2018). Despite these concerns, no research has compared strength between middle-aged adults with or without CAI. This study aimed to compare isometric ankle, knee and hip peak torque in young and middle-aged adults with CAI, copers and un-injured controls. The most important finding from this study was, regardless of age, participants with CAI had decreased sagittal plane ankle and hip isometric peak torque compared to the control group, but not copers.

Early cross-sectional studies mainly examined frontal plane ankle strength as it relates to CAI (Lentell et al., 1995; Wilkerson, Pinerola, & Caturano, 1997; Willems, Witvrouw, Verstuyft, Vaes, & De Clercq, 2002). This was done using various isometric and isokinetic methods to assess torque production in isolation or as a ratio between inversion and eversion. Inconsistent findings from these early studies led to the interpretation that frontal plane strength deficits are not highly correlated with CAI (Kaminski & Hartsell, 2002). Since these initial reports, more recent research has demonstrated reduced isometric (Fraser et al., 2018), concentric (Gribble & Robinson, 2009; Hubbard, Kramer, Denegar, & Hertel, 2007) and eccentric (Fox, Docherty, Schrader, & Applegate, 2008) strength of the sagittal plane ankle movers in individuals with CAI. Our data support these findings as we observed participants with CAI had decrease dorsiflexion and plantar flexion isometric peak torque compared to un-injured controls, but not copers. Applying our results to those previously demonstrated may suggest that sagittal plane ankle strength is a more significant contributing factor to CAI than that of the frontal plane ankle musculature. However, we did not measure inversion or eversion isometric peak torque in the current study and, therefore, cannot rule out the possibility that our participants may have also presented with frontal plane muscle weakness. Nevertheless, reduced sagittal and/or frontal plane ankle strength likely plays a prominent role in the abnormal dynamic movement patterns observed in individuals with CAI (Chinn, Dicharry, & Hertel, 2013; Kipp & Palmieri-Smith, 2012; Koldenhoven, Hart, Saliba, Abel, & Hertel, 2019). Thus, future research and clinicians should continue to assess both frontal and sagittal plane ankle strength in individuals with CAI.

A growing number of studies have started to examine hip and knee strength in individuals with CAI (Gribble & Robinson, 2009; McCann et al., 2018; McCann et al., 2017). Our results showed individuals with CAI have decreased isometric hip extension peak

Table 1
Group means $\pm$ standard deviations or frequency (percentage) for all inclusion criteria, injury history and participant characteristics.

	Younger-Aged Adults			Middle-Aged Adults		
	CAI n = 35	Coper n = 27	Control n = 37	CAI n = 16	Coper n = 16	Control n = 25
All	5.6 $\pm$ 1.6	2.4 $\pm$ 0.7	0.0 $\pm$ 0.0	6.3 $\pm$ 1.4	2.1 $\pm$ 0.9	0.0 $\pm$ 0.0
IdFAI	17.7 $\pm$ 3.8	5.9 $\pm$ 4.4	0.0 $\pm$ 0.0	16.9 $\pm$ 4.4	5.1 $\pm$ 2.9	0.0 $\pm$ 0.0
# of LAS	2.3 $\pm$ 1.8	1.2 $\pm$ 0.4	0.0 $\pm$ 0.0	4.2 $\pm$ 4.7	2.0 $\pm$ 2.3	0.0 $\pm$ 0.0
First LAS (years)	8.5 $\pm$ 7.5	9.3 $\pm$ 6.5	0.0 $\pm$ 0.0	27.1 $\pm$ 11.7	39.1 $\pm$ 42.5	0.0 $\pm$ 0.0
Most Recent LAS (months)	47.5 $\pm$ 67.6	73.8 $\pm$ 68.1	0.0 $\pm$ 0.0	166.2 $\pm$ 158.6	206.5 $\pm$ 159.4	0.0 $\pm$ 0.0
# of "Giving Way" last 6 months	4.2 $\pm$ 4.7	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	5.1 $\pm$ 7.1	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Duration of Symptoms (years)	6.6 $\pm$ 5.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	18.3 $\pm$ 13.9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Modify Activity (yes (%))	16 (43%)	0 (0%)	0 (0%)	5 (31%)	15 (93%)	0 (0%)
Feel at Risk of Re-Injury (yes (%))	27 (73%)	2 (7%)	0 (0%)	13 (81%)	4 (25%)	0 (0%)
Received Rehabilitation (yes (%))	15 (40%)	12 (44%)	0 (0%)	4 (25%)	2 (12%)	0 (0%)

Abbreviations: CAI = chronic ankle instability; All = Ankle Instability Instrument; IdFAI = Identification of Functional Ankle Instability; LAS = Lateral Ankle Sprain.

Table 2Group means $\pm$ standard deviations for all key participant demographic information. Statistical significance is denoted in bold.

	Younger-Aged Adults			Middle-Aged Adults			Injury Main Effect		Age Main Effect		Interaction	
	CAI n = 35	Coper n = 27	Control n = 37	CAI n = 16	Coper n = 16	Control n = 25	F	p-value	F	p-value	F	p-value
Age (years)	26.2 $\pm$ 6.7	25.0 $\pm$ 5.5	26.2 $\pm$ 5.5	50.2 $\pm$ 9.6	54.5 $\pm$ 9.0	56.7 $\pm$ 10.0	2.38	0.096	502.2	<0.001	2.69	0.071
Height (cm)	169.1 $\pm$ 10.4	168.1 $\pm$ 8.5	169.7 $\pm$ 9.1	166.1 $\pm$ 10.2	169.0 $\pm$ 11.4	166.5 $\pm$ 12.7	0.10	0.905	1.05	0.307	0.54	0.582
Mass (kg)	75.2 $\pm$ 15.9	68.1 $\pm$ 13.5	70.2 $\pm$ 14.6	81.7 $\pm$ 17.5	78.3 $\pm$ 16.4	72.2 $\pm$ 12.4	3.14	0.046	6.16	0.014	0.93	0.396
BMI (kg/m ²)	26.1 $\pm$ 4.2	24.0 $\pm$ 4.1	23.9 $\pm$ 3.4	30.0 $\pm$ 7.9	27.4 $\pm$ 4.7	26.4 $\pm$ 5.5	5.14	0.007	16.14	<0.001	0.27	0.757

Abbreviations: **CAI** = chronic ankle instability; **cm** = centimeters; **kg** = kilograms; **kg/m²** = kilograms/square meter.**Table 3**Group means $\pm$ standard deviations for all isometric strength outcomes variables. Statistical significance is denoted in bold.

	Younger-Aged Adults			Middle-Aged Adults			Injury Main Effect		Age Main Effect		Interaction	
	CAI n = 35	Coper n = 27	Control n = 37	CAI n = 16	Coper n = 16	Control n = 25	F	p-value	F	p-value	F	p-value
Dorsiflexion (Nm/kg)	0.31 $\pm$ 0.08	0.43 $\pm$ 0.10	0.41 $\pm$ 0.12	0.27 $\pm$ 0.09	0.27 $\pm$ 0.09	0.31 $\pm$ 0.08	6.11	0.003	33.6	<0.001	3.49	0.033
Plantarflexion (Nm/kg)	0.57 $\pm$ 0.16	0.61 $\pm$ 0.16	0.71 $\pm$ 0.18	0.49 $\pm$ 0.16	0.57 $\pm$ 0.13	0.56 $\pm$ 0.12	4.86	0.009	10.6	0.001	1.51	0.223
Knee Extension (Nm/kg)	2.09 $\pm$ 0.54	2.34 $\pm$ 0.59	2.50 $\pm$ 0.68	1.96 $\pm$ 0.71	2.05 $\pm$ 0.53	1.85 $\pm$ 0.43	0.97	0.380	11.2	0.001	2.29	0.104
Hip Abduction (Nm/kg)	1.72 $\pm$ 0.52	1.99 $\pm$ 0.45	2.04 $\pm$ 0.59	1.60 $\pm$ 0.63	1.68 $\pm$ 0.48	1.63 $\pm$ 0.31	1.72	0.182	10.3	0.002	0.99	0.372
Hip Extension (Nm/kg)	1.65 $\pm$ 0.54	2.03 $\pm$ 0.55	2.12 $\pm$ 0.58	1.32 $\pm$ 0.44	1.37 $\pm$ 0.63	1.53 $\pm$ 0.75	4.11	0.018	27.8	<0.001	0.91	0.403

Abbreviations: **CAI** = chronic ankle instability; **Nm/kg** = Newton meter/kilogram.

torque compared to controls, but not copers. This finding lends further support to those by McCann et al. (McCann et al., 2018; McCann et al., 2017), who used similar methods as the current study and documented reduced hip extension and abduction strength in people with CAI compared to copers and un-injured controls. In contrast, we did not observe a between group difference for isometric knee extension strength. This differs from a previously reported observation of reduced concentric knee strength individuals with CAI and un-injured controls (Gribble & Robinson, 2009). Interestingly, Gribble & Robinson (Gribble & Robinson, 2009) did not observe a difference between un-injured controls and CAI participants for concentric hip flexion or hip extension peak torque. The difference in the methodology used between studies (e.g., isometric vs. concentric) might explain the varying results. Future research should consider the mode of contraction, measurement device or body position when assessing lower extremity muscle strength associated with CAI to increase homogeneity across studies and allow pooling of results.

While the current study observed the average sagittal plane ankle and hip isometric peak torque was lower in participants with CAI relative to the un-injured controls, no statistical difference was found comparing CAI and copers for any strength measure. Similarly, no difference was noted between copers and un-injured controls. These findings are in agreement with previous studies comparing ankle and hip strength between individuals with CAI, copers and un-injured controls (Fraser et al., 2018; McCann et al., 2019). Prospective studies have found that lower ankle (Baumhauer et al., 1995; Fousekis et al., 2012) and hip (De Ridder, Witvrouw, Dolphens, Roosen, & Van Ginckel, 2017; Powers, Ghoddosi, Straub, & Khayambashi, 2017) strength are intrinsic risk factors for sustaining a non-contact lateral ankle sprain in youth and college-aged athletes. This may explain why the average strength for the coper group, regardless of age, tended to be lower than that of the un-injured controls, but higher than participants with CAI (dorsiflexion [Nm/kg] – CAI: 0.30 $\pm$ 0.08, coper: 0.37 $\pm$ 0.12, control: 0.37 $\pm$ 0.12; plantar flexion [Nm/kg] – CAI: 0.55 $\pm$ 0.16, coper: 0.59 $\pm$ 0.15, 0.65 $\pm$ 0.18; hip extension [Nm/kg] – CAI: 1.56 $\pm$ 0.53, coper: 1.78 $\pm$ 0.66, control: 1.89 $\pm$ 0.71). However, this trend did not arise to be statistically different between groups. Therefore, our results suggest that greater deficits in sagittal plane

ankle and hip peak torque contribute more to the self-reported chronic dysfunction and episodes of 'giving way' commonly associated with CAI. Examining the relationship between lower extremity strength and patient-reported outcomes is needed to confirm this theory and would be an area for future research to investigate.

The unique aspect of this study is that we enrolled young and middle-aged adults with CAI, copers and un-injured controls. The decreased strength in middle-aged participants compared to young adults, regardless of injury history, was expected. Longitudinal research suggests lower extremity muscle strength peaks around the second or third decade of life and remains unchanged until the fifth decade. In the fifth decade, strength begins to decline at a rate of 10% to 15% per decade (Hughes et al., 2001; Hurley, 1995). With this known decline in strength combined with the already reduced strength observed in young adults with CAI, we hypothesized that middle-aged participants with CAI would have greater deficits in strength compared to all other groups. A significant interaction between Injury and Age was observed for only dorsiflexion isometric peak torque. This interaction effect demonstrated that among the un-injured control and coper groups, dorsiflexion isometric strength decreased with advancing age. However, there was not a difference in dorsiflexion isometric strength between young and middle-aged adults with CAI. This finding suggests that young adults with CAI already present with similar dorsiflexion strength as their middle-aged counterparts. Likewise, the between-group differences described above for plantar flexion, dorsiflexion and hip extension provide further support that the strength deficits observed in young adults persist into middle adulthood and do not continue to decline. Therefore, our results add to the current body of literature surrounding lower extremity muscle strength associated with CAI and age (Kosik et al., 2019; Simon & Docherty, 2018).

The chain of events that lead an individual with CAI to present with diminished strength throughout the lower extremity is currently unknown and the cross-sectional design of our study limits us from further elucidating the mechanism(s). One possible scenario is that lower extremity strength is not a direct consequence of CAI, but rather a risk factor for an ankle sprain. Prospective studies found decreased hip extension and hip abduction strength as a risk factor for an ankle sprain in soccer athletes (De

Ridder et al., 2017; Powers et al., 2017). Thus, we cannot rule out the chance that the strength deficits observed in the current study were not already present when our participants developed CAI. Other researchers (DeJong et al., 2020; Fraser et al., 2018; Gribble & Robinson, 2009) have hypothesized the diminished lower extremity strength among individuals with CAI is related to peripheral and central neural alterations previously demonstrated (Kosik, Terada, Drinkard, McCann, & Gribble, 2017; McVey, Palmieri, Docherty, Zinder, & Ingersoll, 2005; Sedory, McVey, Cross, Ingersoll, & Hertel, 2007). Another possible explanation might be related to the presence of chronic ankle pain (Al Adal, Pourkazemi, Mackey, & Hiller, 2019), fear of re-injury (Houston, Van Lunen, & Hoch, 2014) or increased self-reported functional limitations (Houston, Hoch, & Hoch, 2016) causing people with CAI to become less physically active (Hubbard-Turner & Turner, 2015). There are numerous benefits of participating in regular physical activity such as maintaining proper body weight, reducing risk for cardiovascular disease, and strengthening bones and muscles. Therefore, the decreased physical activity shown to occur early in life (Hubbard-Turner & Turner, 2015) may have a global impact on muscle strength and other outcomes associated with CAI. An animal model has also shown that mice induce with an ankle sprain decrease their physical activity throughout their lifespan (Hubbard-Turner, Wikstrom, Guderian, & Turner, 2015). Although we did not record physical activity levels, it's possible that avoiding physical activity into middle adulthood might contribute to the persistent strength deficits observed in the current study. Therefore, educating young adults with CAI on the importance of being physically active and maintaining a healthy lifestyle as they get older might be a topic clinicians and researchers need to consider for this patient population and future investigations. Clearly, all of this remains speculative at the moment and numerous research studies are needed to corroborate any of these hypotheses because the underlying relationship between CAI and lower extremity muscle strength is likely multi-factorial and does not rely on a singular explanation.

Nevertheless, the reduced ankle and hip strength observed by the CAI group in the present study is a concern because it can impact functional outcomes. For example, lower isometric hip strength is associated ($R^2 = 0.25$) worse dynamic balance in young adults with CAI (McCann et al., 2017). Lower hip strength has also been shown to be an independent predictor for an ankle sprain in youth soccer players (De Ridder et al., 2017). However, the exact impact strength has on middle-aged adults with CAI is less clear because no investigation currently exists among this relationship among this age cohort. It's reasonable to speculate that muscle strength can also translate into worse functional outcomes. This is supported by previous research showing a relationship between measures of isometric plantarflexion and hip strength and balance and gait in otherwise healthy middle-aged adults (Holviala, Sallinen, Kraemer, Alen, & Hakkinen, 2006; Izquierdo, Aguado, Gonzalez, Lopez, & Hakkinen, 1999; Muehlbauer, Gollhofer, & Granacher, 2012). Moreover, while frontal plane strength was not assessed in the current study, it's possible inversion or eversion strength deficits may also be present in middle-aged adults with CAI and have an impact on functional outcomes.

4.1. Limitation

This investigation does have limitations that need to be considered. A primary limitation of this study is the retrospective design prevents the ability to determine whether lower levels of strength are the result of cause of CAI. The design of this study also limits our understanding of knowing how long individuals with CAI have had strength deficits. Related to this, the exact timepoint any participant developed CAI is unknown. We aimed to identify

participants who had developed CAI by the time they reached the end of early adulthood by requiring participants to self-report their initial ankle sprain prior to 40 years of age. The average duration each participant experienced symptoms of CAI (e.g. 'giving way' or instability) provides supporting evidence that they developed as a young adult. Lastly, there are currently no established guidelines for identifying middle-aged individuals with CAI. The current recommendations are based on instruments designed and validated to discriminate between young adults with and without CAI (Gribble et al., 2014). It is possible that middle-aged adults may experience different symptoms associated with CAI. This could have led to some participants enrolled to appear as a copers. Despite this concern, all participants allocated to the CAI group scored beyond the established thresholds on the AII and IdFAI. Other limitations of our study include only unilateral strength assessments were recorded; participants did not take part in a familiarization session before participating and were given a single practice trial; only isometric strength was measured; no functional outcomes were recorded; and we did not match for sex, injury limb or age. Future work should aim to examine the impact of age and strength on functional outcomes during dynamic activities.

5. Conclusion

Isometric strength is a common method used by clinicians to measure a patients' level of strength after a joint injury and as an indicator to determine progression during rehabilitation. The results from the current study indicate that, regardless of age, participants with CAI had reduced ankle and hip isometric peak torque compared to un-injured controls, but not copers. These results provide evidence that highlight the need for clinicians to assess potential strength throughout the lower extremity when identifying specific impairments in patients with CAI.

Ethical approval

This study was approved by the University of Kentucky Institutional Review Board ((IRB# 15-0966-P1H) and each participant signed an approved consent document prior to enrollment.

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Declaration of competing interest

None declared.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ptsp.2020.02.014>.

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